When do students miss school? A data analysis and machine learning study of attendance in two classes

Kessia MK, Class XII, August 2026

Summary

School attendance registers are kept every day but rarely analysed. This study used one year (April 2025 to April 2026) of the daily register of two classes, Class VIII (31 students) and Class IX (27 students), of a school in Coimbatore, Tamil Nadu. Students appear only as numbers. Class IX attendance was 95.52% and Class VIII was 87.63% (difference 7.89 percentage points, $p < 0.001$); 19 of 31 Class VIII students and none in Class IX were below 90%. Weekday and rain made no clear difference. Class VIII attendance fell to 76.34% in the three days after examinations, but almost entirely after the December and final examinations, which end next to long holidays. K-means clustering found a group of 15 Class VIII students with 82.6% attendance. A simple rule (10% absence in the first four weeks) identified 15 of the 20 students who were later chronically absent, with 74.2% accuracy against a baseline of 64.5%.

1. Introduction

Missing school matters most when it happens close to a test. Gottfried and Kirksey (2017) found that absences in the 30 days before a state test were linked to lower scores in elementary students. Balfanz et al. (2007) described attendance as one of the early signals used to identify middle-grade students who are drifting away from school.

In Tamil Nadu, the School Education Minister stated in 2023 that students need 75% attendance to be allowed to write the public examinations (DT Next, 2023). Researchers in other countries use a stricter line: a student who misses 10% or more of school days, that is, whose attendance is below 90%, is called chronically absent (America's Health Rankings, 2024). Both lines are used in this study.

Patterns in absence also depend on the calendar. Clifton-Sprigg and James (2025) found that absence in England was about 20% higher on Fridays than on other weekdays. This study asks whether similar calendar patterns appear in an Indian school, and whether a simple analysis of the register can show which students are at risk.

The questions were: (1) How different is attendance in Class VIII and Class IX? (2) Does attendance change with the month, the weekday, rain or examinations? (3) Can machine learning group students by their pattern of absence? (4) Can the first four weeks of the year show who will be chronically absent later?

2.Data and methods

2.1 Data and ethics

The school shared its daily register for each class (all sections combined), its holiday list, the examination dates and the daily rainfall. Each student was marked present (P) or absent (A) on every working day. The school had already replaced names with numbers, and no personal details were used. The register is published with the code as attendance_register.csv, with students shown only as numbers. Only group-level results are

reported in this write-up. The school principal gave the permission for use of the data for the analysis.

Table 1 describes the data. Class IX worked on second Saturdays and Class VIII did not, so every result is given as a percentage of the days that class worked.

	Class VIII	Class IX
Students	31	27
Working days	244	253
Student-days (students × days)	7,564	6,831
Absences	936	306
Attendance (% of student-days)	87.63	95.52
Days absent per student (average)	30.2	11.3
Students below 90% (chronically absent)	19 of 31	0 of 27
Lowest and highest student attendance (%)	73.77 to 96.72	92.89 to 98.42

Table 1. The two classes, April 2025 to April 2026.

2.2 Measures

Attendance (%) = days present ÷ working days × 100. A rainy day was a day with 2.5 mm of rain or more (India Meteorological Department definition). Around each examination, the five working days before it, the exam days, and the three working days after it were compared with all other (normal) days. The first working day after a break that contained a listed holiday was called an after-holiday day.

2.3 Statistical tests

The difference between the classes was tested by shuffling the class labels 10,000 times (permutation test). For comparisons inside a class (exam days against normal days, rainy against dry days, Monday, Friday and Saturday against other days) each student's own attendance was compared on the two kinds of day. The mean difference is given with a 95% range from 10,000 resamples of students and a p-value from random sign flips of the paired differences. Intervals for proportions are Wilson 95% intervals.

2.4 Machine learning

K-means clustering grouped students using four measures from the register: absence for the year, the change in absence between the first and second halves of the year (split at 15 October 2025), the average length of an absence spell, and absence on after-holiday days. The measures were standardised, 2 to 6 groups were tried, and the number with the highest silhouette score was kept (random seed 42).

For early warning, Class VIII was used because no Class IX student fell below 90%. A student was flagged if absent on 10% or more of the first 22 working days (four school weeks). The outcome was attendance below 90% in the rest of the year. The rule was compared with logistic regression (each student left out in turn) and with a baseline that gives every student the commonest answer.

2.5 Software

The analysis was written in Python with pandas, NumPy, matplotlib and scikit-learn (Pedregosa et al., 2011). The code and the register (attendance_register.csv) are on GitHub: https://github.com/kessiamk/school-attendance-study

3.Results

3.1 The two classes

Class IX attendance was 95.52% and Class VIII was 87.63%, a gap of 7.89 percentage points ($p < 0.001$). A Class IX student missed about 11 days in the year and a Class VIII student about 30. Nineteen of the 31 Class VIII students (61%) were below 90%, and two were below 75%. In Class IX the lowest student attendance was 92.89%.

3.2 Month, weekday and rain

Class IX attendance stayed between 94% and 97% in most months, with a dip in August 2025 (92.75%) (Figure 1). Class VIII was highest in July 2025 (90.20%) and lowest in January 2026 (85.41%) and April 2026 (83.41%, only seven working days). From April to December 2025 Class VIII attendance was 88.21%; from January to April 2026 it was 86.26%. For Class IX the same periods gave 95.07% and 96.66%.

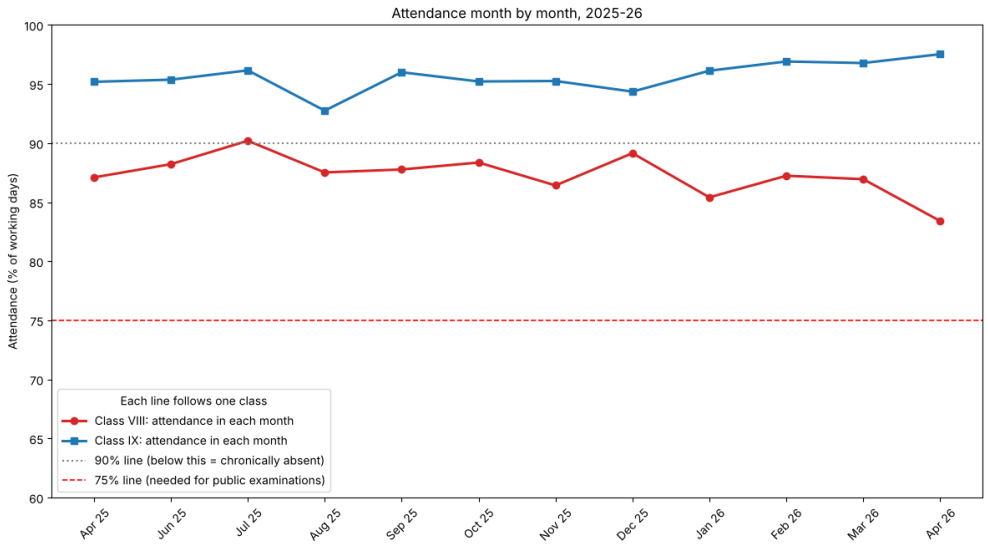


Figure 1. Attendance in each month. Red line, Class VIII; blue line, Class IX. The dotted line marks 90% and the dashed line 75%. April 2026 had only 7 (Class VIII) and 3 (Class IX) working days.

The weekday made little difference. Class VIII ranged from 86.50% (Wednesday) to 88.67% (Monday), and Class IX from 94.85% (Friday) to 96.55% (Wednesday). Monday, Friday and Saturday together were not lower than other days (Class VIII: −0.05 percentage points, 95% range −1.33 to 1.25; Class IX: −0.63, −1.71 to 0.42). Rain made no clear difference either. There were 40 rainy school days. Class IX attendance was 95.28% on rainy days and 95.57% on dry days; Class VIII was 88.31% and 87.49% (Class VIII difference +0.81, −1.09 to 2.62; Class IX −0.29, −1.61 to 0.92).

3.3 Examinations

Attendance on exam days was high in both classes: 97.42% in Class VIII and 97.94% in Class IX (Figure 2). For Class VIII this was 10.17 points above normal days (95% range 7.42

to 12.96); for Class IX it was 2.73 points above (1.29 to 4.04). In the three days after an exam, Class VIII attendance fell to 76.34%, which is 10.91 points below normal days (−15.61 to −6.57). Class IX showed no fall (95.29% against 95.21% on normal days).

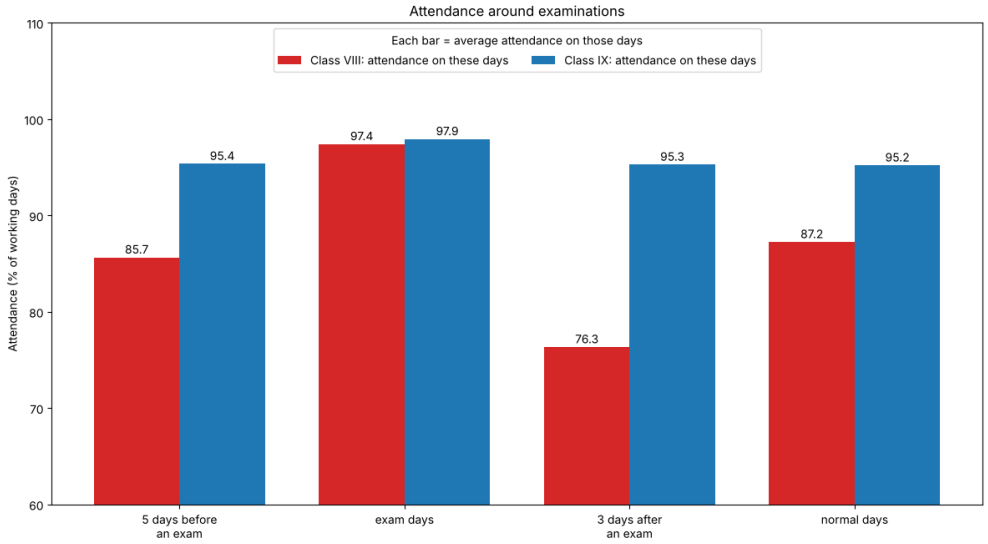


Figure 2. Average attendance in the five days before an exam, on exam days, in the three days after an exam, and on normal days. Red bars, Class VIII; blue bars, Class IX.

The fall was not the same after every exam (Table 2). It was small after Exam 1 in July and large after the first-semester exam in December and the final exam in April. The lowest attendance of the year for Class VIII was on 27 December 2025 and 8 April 2026, when only 51.61% of the class was present on each day.

Exam	Class VIII (%)	Class IX (%)
Exam 1 (July)	88.17	97.53
Exam 2 (September or October)	81.72	93.83
First semester (December)	66.67	92.59
Final (March to April)	68.82	98.15

Table 2. Attendance in the three working days after each examination. Normal days: 87.25% (Class VIII) and 95.21% (Class IX).

3.4 Is absence a steady habit?

Each student's attendance before and after 15 October 2025 was compared. In Class VIII the correlation was r = 0.82, so students who missed many days early also missed many later. In Class IX it was r = 0.00, because almost every student had similar, high attendance.

3.5 Groups found by k-means

Two groups gave the cleanest separation (silhouette 0.488; 0.456 for three groups). Group A had 43 students (16 from Class VIII and all 27 from Class IX) with average attendance of 94.3%, and Group B had 15 students, all from Class VIII, with 82.6% (Table 3, Figure 3). Group B students were absent on 24.6% of after-holiday days, against 5.4% in Group A. Their absence spells were not much longer (1.3 days against 1.1 days); they miss school often, in short spells, and especially after breaks.

Group	Students	Class VIII	Class IX	Attendance (%)	Absent on after-holiday days (%)
A	43	16	27	94.3	5.4
B	15	15	0	82.6	24.6

Table 3. The two groups found by k-means.

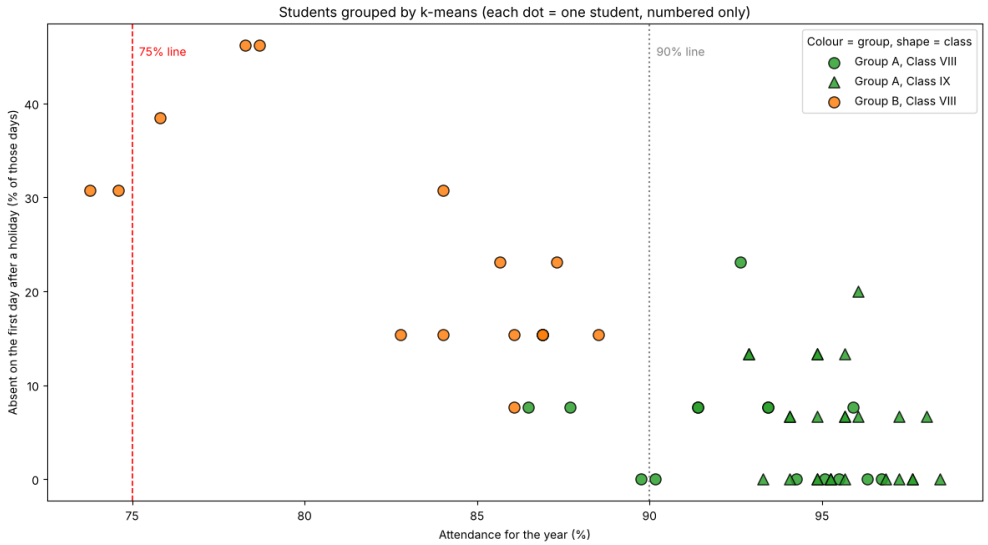


Figure 3. Each dot is one student, shown only as a dot. Green, Group A; orange, Group B. Circles are Class VIII and triangles are Class IX. Dots further right attended more; dots higher up were absent more often on the first day after a holiday.

3.6 Early warning from the first four weeks

Twenty of the 31 Class VIII students had attendance below 90% in the rest of the year. The rule flagged 18 students; 15 of them became chronically absent and 3 did not, and 5 students who were not flagged also became chronically absent (Figure 4). The rule was right for 23 of 31 students (74.2%, 95% range 56.8% to 86.3%). Giving every student the commonest answer would be right for 64.5%, so the range is wide enough to include the baseline. Of the flagged students 83% (95% range 61% to 94%) became chronically absent, against 38% (5 of 13) of those not flagged. Early and later absence were correlated (r = 0.65). Logistic regression, tested by leaving each student out in turn, was right for 64.5% of students, the same as the baseline.

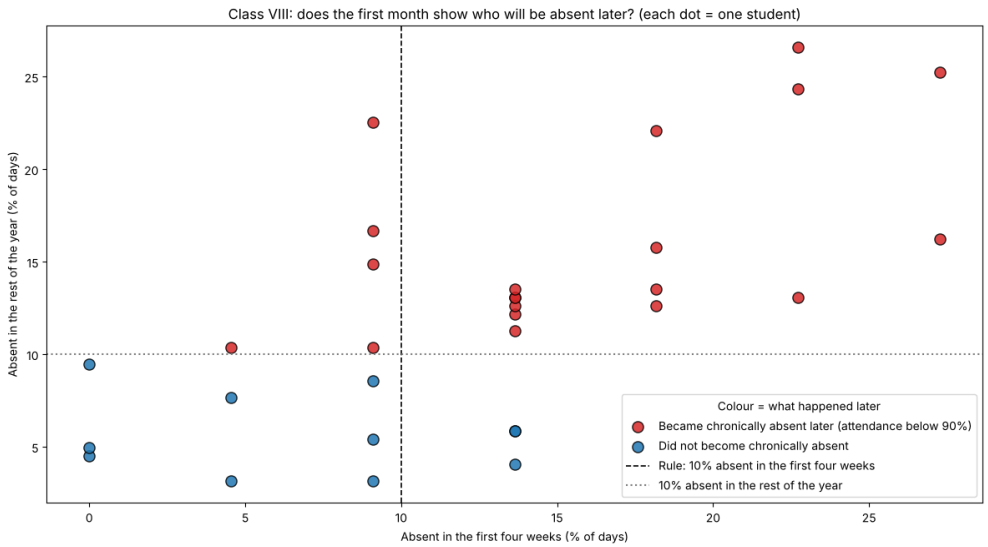


Figure 4. Class VIII. Each dot is one student. Red dots became chronically absent later (below 90% attendance); blue dots did not. The dashed vertical line is the 10% rule for the first four weeks.

3.7 Hypotheses

Hypothesis	Result	Evidence
H1. The two classes differ	Supported	7.89 points, p < 0.001
H2. Lower attendance on Monday, Friday, Saturday	Not supported	Class VIII −0.05 (−1.33 to 1.25)
H3. Lower attendance on rainy days	Not supported	Class VIII +0.81 (−1.09 to 2.62)
H4. Higher attendance on exam days	Supported	Class VIII +10.17, Class IX +2.73
H5. Lower attendance after exams	Class VIII only, mainly after two exams	−10.91 (−15.61 to −6.57); Table 2
H6. The first four weeks show who will be chronically absent	Partly	74.2% against 64.5% baseline; range 56.8% to 86.3%

Table 4. Result for each hypothesis. Differences are in percentage points with 95% ranges.

4.Discussion

The clearest result is the gap between the classes and the concentration of absence in a group of students. Absence in Class VIII is a habit of particular students: attendance in the first and second halves of the year was strongly correlated (r = 0.82), and one group of 15 students accounts for much of the difference. Because Class VIII and Class IX are different students, this study cannot say whether the gap comes from the class level or from the students.

The weekday pattern found in England (Clifton-Sprigg and James, 2025) did not appear here. The weekday differences were about 2 percentage points or less, and the intervals for Monday, Friday and Saturday included zero. Rain also made no clear difference; the heaviest school-day rainfall was 29.4 mm.

The fall in Class VIII attendance after exams needs care. It was large after the December and final exams (66.67% and 68.82%) and small after Exam 1 (88.17%). The December exam ended on 24 December, and Christmas, New Year's Day and several days without

school lay close behind it. The final exam ended on 7 April, and the register ends three days later, so the three days after it were the last days of the school year. After Exam 2 (81.72%), Deepavali fell about a week later. Clifton-Sprigg and James (2025) also link absence to holidays: Friday absence in England was more common in the weeks leading up to bank holidays and half-term breaks. The data here cannot separate two explanations: students treating the days after an exam as a holiday, and students extending a long break. The k-means result points the same way, because Group B students were absent on a quarter of after-holiday days.

For the school, three practical steps follow. The attendance percentage at the end of the first month is a cheap warning sign, because 83% of students flagged by the 10% rule were chronically absent later. The days after the December and final exams could be planned with activities and a message to parents that they are full school days. Students in Group B need a conversation about the reason for absence, which the register does not record.

A machine learning model did not beat a plain rule. With 31 students the logistic regression had too few examples, and its accuracy (64.5%) equalled the baseline. This is a result in itself: a simple threshold on the attendance percentage is easier to explain and was at least as good. With more years of data I would train the rule on one year and test it on the next, which is a fairer test than 31 students in one year. With the reason for each absence (illness, family travel or a school event), I could separate absences that a school can reduce from those it cannot, and check whether the extra absence after December and final exams comes from students extending holidays.

5.Limitations

- Only one year of two classes from one school was studied, so results may not apply to other schools.
- The register shows only present or absent. It does not show why a student was absent, for example illness or a school event.
- Class VIII and Class IX are different students, so a class effect and a student effect cannot be separated.
- The exam and holiday effects overlap in the calendar, so the fall after exams cannot be fully separated from the effect of long breaks.
- With 31 students the early-warning result has a wide range (56.8% to 86.3%) that includes the baseline accuracy of 64.5%.
- The k-means groups depend on the four measures chosen. Group B is a pattern of attendance, not a judgement about any student.
- Examination dates were known only as periods, not as the date of each paper.

Acknowledgements

Mrs. Archana Jinesh, my computer science teacher, gave guidance on this project. The school shared its attendance data with permission. AI tools helped write and check the Python code and drafted the first version of this write-up. I ran the code, checked the main results against the register, and rewrote the text in my own words.

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